DIARRHEA

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Definition :

- Stool water content >200 mL/24 hours.
- According to Bristol stool chart :
  - Type 1 & 2 : Constipation.
  - Type 6 & 7 : Diarrhea.



Bristol stool chart

Physiology of absorption :

1. Luminal.
2. Mucosal :
   - Small intestine (majority).
   - maximum absorption.
3. Post mucosal.

macronutrients for absorption :

- Fat :
  - most calorie dense nutrient.
  - most specific nutrient affected in malabsorption.
- Carbohydrates.
- Proteins.

Frequency Classification

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	Acute	Persistent	Chronic
Duration	<2 weeks (Nuisance symptom)	>2 weeks	>4 weeks

ETIOLOGY

Acute diarrhea :

Infectious : Viral infection (m/c).

- m/c in adults : Norovirus.
- m/c in children : Rotavirus.
- Self-limiting.

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Chronic diarrhea :

D/d	Etiology
Osmotic	malabsorption
Secretory	• Tumor • Toxin
Inflammatory	Inflammatory bowel disease

Note :

Malassimilation :

- Maldigestion + malabsorption.
- Diminished intestinal digestion/absorption of one or more nutrients.

Pathological Classification 00:10:13

1. OSMOTIC DIARRHEA

Most consistent clinical symptom of malabsorption.

Steatorrhea :

Hallmark of malabsorption.

Definition : Passing of pale, bulky, fatty, malodorous, greasy stools.

Clinical presentation : Diarrhea (most consistent clinical finding).

Pathology :

↓ fat absorption in small intestine

↓ In colon

Unabsorbed macronutrients ↑ stool osmolality (osmotically active)

↓

Stool draws water from large intestinal epithelial cells

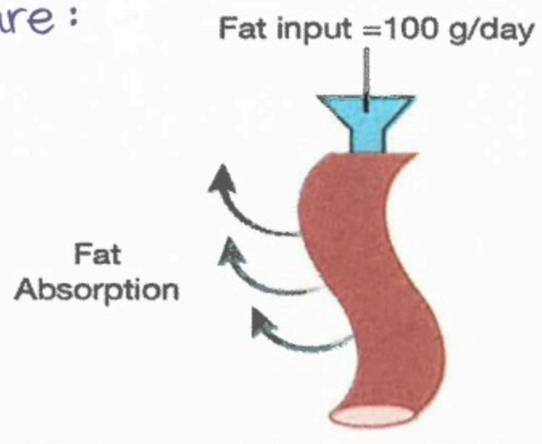
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Osmotic diarrhea

Investigations :

1. 72 hour fecal fat test : Gold standard.

   Procedure :



Normal : <7 g/day

- >7 g/day for 3 days (D3, D4, D5)
- Stool fat excretion ≥7 %

} Steatorrhea

2. Qualitative stool fat :
   - Sudan III.
   - Sudan IV.
   - Oilred O.

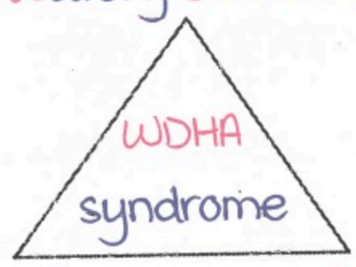
2. SECRETORY DIARRHEA

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Etiology :

a. Toxins → ETEC, V. cholera (heat labile toxin)
a. Toxins → Enteropathogenic virus (rotavirus)

b. Tumours : VIPoma → Watery Diarrhoea



Hypokalemia
Acidosis

Note : No structural damage seen.

	Osmotic diarrhoea	Secretory diarrhoea
Stool osmotic gap	Increased (>100 mOsm/kg)	Normal/decreased (< 50 mOsm/kg). ↑ stool $Na^+ + K^+$ (By toxins).
Response to fasting	Improvement	No change
Stool pH	<5.5 (Fermentation of unabsorbed carbohydrates → Acids)	>6.0

Note : Stool osmotic gap

- measured osmolality (osmometer) − calculated osmolality
  → 290 − 2 (Stool $Na^+ + K^+$)
- Normal : 50–100 mOsm/kg.

3. FACTITIOUS DIARRHOEA

Clinical presentation :

- Adolescent female.
- Chronic diarrhea.
- Fatigue.
- Weight loss.

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Findings :

Stool collection	Stool osmolality
Unsupervised (mixed with water)	Very low (~ 16 mOsm/kg)
Supervised	Normal (~ 279)

Anatomical Classification

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SMALL INTESTINAL V/S LARGE INTESTINAL DIARRHEA

		Small intestinal diarrhea	Large intestinal diarrhea
Etiology		• Osmotic. • Secretory : – ETEC. – Vibrio cholera.	• Infections : Invasive organisms (Shigella). • Inflammation : Ulcerative colitis.
Clinical presentation		Large (↓ nutrient absorption)	Small
Diarrhoea	Volume		
	Consistency	Watery	Watery
	Frequency/ urgency	↓	↑
	Pus/blood/ mucus	Absent	Present
	Tenesmus	Absent	Present
	Dyschezia	Absent	Present
Abdominal pain		Absent	Present
Fever		Absent	Present
Abdominal cramps		Present	Present
Bloating		Present	Present
Weight loss		Present (If persistent)	Rare
Vomiting		Maybe present	Rare
Steatorrhoea		Present	Absent

APPROACH TO DIARRHEA

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